

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Multimeter, Part Number 3164489.

Additional Service Items

- No additional service items required.

General Information

The transmission gear oil temperature sensor is used to measure the temperature of the transmission oil and is located on the marine gear oil cooler.

Note : The content of this procedure is for the Cummins® supplied sensor. The transmission gear oil temperature sensor may be supplied by the original equipment manufacturer (OEM), therefore the Cummins®-supplied service tools listed in this procedure may not fit. See equipment manufacturer service information.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

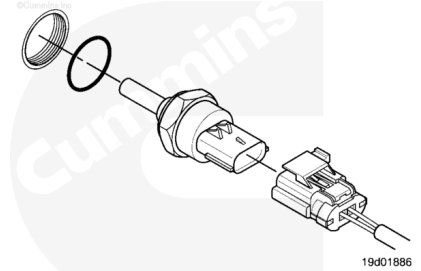
- Disconnect the batteries. See equipment manufacturer service information.
- Clean the area around the sensor.
- Drain the marine gear oil cooler. Use the following procedure in the Marine QSB6.7 CM2250 Service Manual, Bulletin 4310611. Refer to Procedure 008-041 in Section 8.
(/qs3/pubsys2/xml/en/procedures/154/154-008-041-tr.html)

- Drain the marine gear oil cooler. Use the following procedure in the Marine QSL9 M CM2250 L106 Service Manual, Bulletin 4358343. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/263/263-008-041-tr.html>)
- Drain the marine gear oil cooler. Use the following procedure in the Marine QSC8.3 CM850 Troubleshooting and Repair Manual, Bulletin 4021418. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/45/45-008-041-tr.html>)
- Drain the marine gear oil cooler. Use the following procedure in the Marine QSM11 CM570 Service Manual, Bulletin 3666322. Refer to Procedure 008-084 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-084-tr.html>)

Remove

Disconnect the sensor connector from the harness.

Remove the sensor from the marine gear oil cooler.

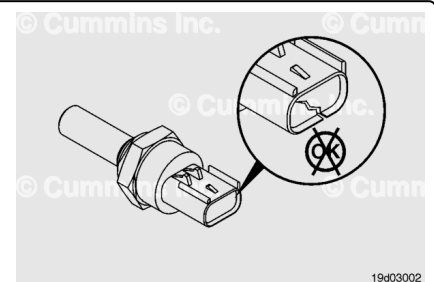


Clean and Inspect for Reuse

Inspect the harness connector and sensor for the following:

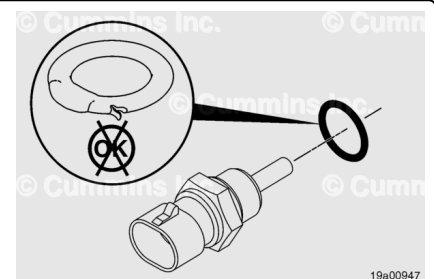
- Cracked or broken connector shell
- Missing or damaged connector seals
- Dirt, debris, or moisture, in or on the connector pins
- Corroded, bent, broken, pushed back, or expanded pins
- Chipped, cracked, extruded or damaged sensor.

Repair or replace parts as necessary.

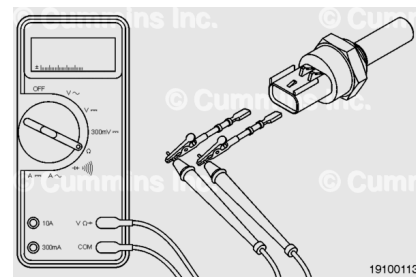


Inspect the sensor for the following:

- Swollen o-ring
- Nicks or cuts in or on the o-ring.



Measure the resistance across the sensor signal and return terminals to be sure they agree with the table shown below. If the sensor ambient conditions change, the resistance values should be measured **only** after 2.5 minutes of the ambient change.



Temperature (°C)	Temperature (°F)	Resistance (ohms)
0	32	30k to 37k
25	77	9.3k to 10.7k
50	122	3.2k to 3.8k
80	176	1.1k to 1.3k
100	212	627 to 648

Install

Check to be sure the sensor has an o-ring installed.

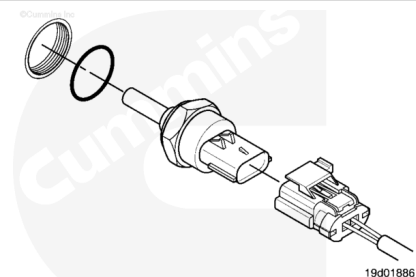
Lubricate the o-ring with clean engine oil before installation.

Install the sensor into the marine gear oil cooler.

Tighten the sensor.

Torque Value: 15 ±3 n·m [133 ±27 in-lb]

Connect the harness connector to the sensor.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Fill the marine gear oil cooler. Use the following procedure in the Marine QSB6.7 CM2250 Service Manual, Bulletin 4310611. Refer to Procedure 008-041 in Section 8. ([/qs3/pubsys2/xml/en/procedures/154/154-008-041-tr.html](http://qs3/pubsys2/xml/en/procedures/154/154-008-041-tr.html))

- Fill the marine gear oil cooler. Use the following procedure in the Marine QSL9 M CM2250 L106 Service Manual, Bulletin 4358343. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/263/263-008-041-tr.html>)
- Fill the marine gear oil cooler. Use the following procedure in the Marine QSC8.3 CM850 Troubleshooting and Repair Manual, Bulletin 4021418. Refer to Procedure 008-041 in Section 8. (</qs3/pubsys2/xml/en/procedures/45/45-008-041-tr.html>)
- Fill the marine gear oil cooler. Use the following procedure in the Marine QSM11 CM570 Service Manual, Bulletin 3666322. Refer to Procedure 008-084 in Section 8. (</qs3/pubsys2/xml/en/procedures/35/35-008-084-tr.html>)
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine. Check for leaks.
- Perform system test to verify proper function. Refer to Procedure 015-047 in Section 15. (</qs3/pubsys2/xml/en/procedures/513/513-015-047.html>)